Gears and wheels

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use gear ratio and wheel circumference to predict an ideal travel speed.

SUCCESS CRITERIA - I can...

1. Connect gear teeth with speed ratio.
2. Calculate distance per wheel revolution.
3. Explain the tradeoff between ideal speed and available torque.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 30-turn coil has flux per turn change by 0.005 Wb in 0.3 s. Find average induced-emf magnitude.

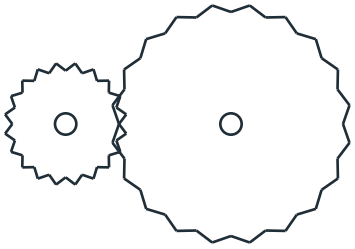
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $rpm_{out} = rpm_{in} \times teeth_{driver}/teeth_{driven}$; $C = \pi d$
$v = C \times rpm_{wheel}/60$.
Ideal gears and rolling without slip; C in meters gives v in m/s. Loading and losses change real performance.

Worked reasoning: A 10-tooth driver turns a 30-tooth driven gear. The driven gear makes one turn for every three driver turns. In an ideal gear pair, slower rotation trades for greater torque.

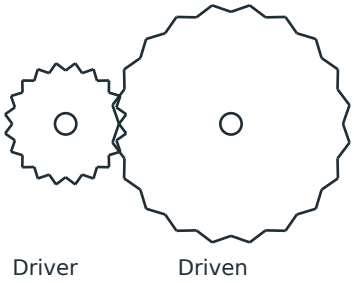
Your turn: Draw both gears, label tooth counts and opposite rotation directions.



Marzano check - circle: ① ② ③

03 Read and interpret the model

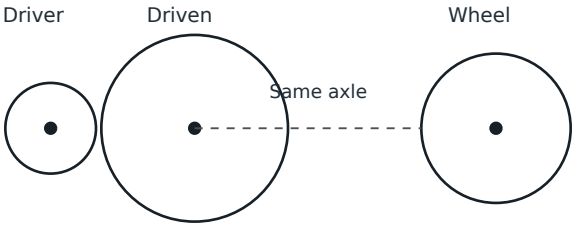
A small driver gear turns a larger driven gear. Predict which turns more slowly and indicate their opposite rotation directions.



Marzano check - circle: ① ② ③

04 Apply the physics

A model-car motor turns at 600 rpm. A 10-tooth driver turns a 30-tooth driven gear attached to a wheel of diameter 0.060 m. Label teeth, wheel size and rotation directions. Calculate wheel rpm and ideal ground speed with no slip.



Label teeth, rotation rates and wheel size.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A team chooses the largest wheels because the formula predicts more distance per turn. Explain why measured acceleration or speed might still decrease.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

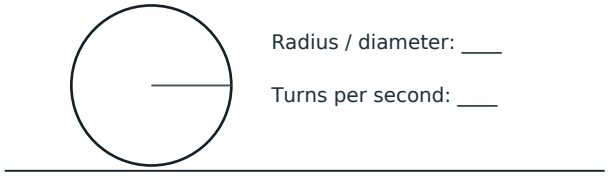
A 7 Ω resistor is in series with two parallel 10 Ω resistors across 24 V. Find parallel equivalent, total resistance and supply current.
Draw the stated circuit and label quantities relevant to the question.

Draw the complete circuit and label shared nodes.

Marzano check - circle: 1 2 3

07 Exit check

A 12-tooth driver turns a 36-tooth gear. Find ideal output torque as a multiple of input torque and output speed as a fraction of input speed.

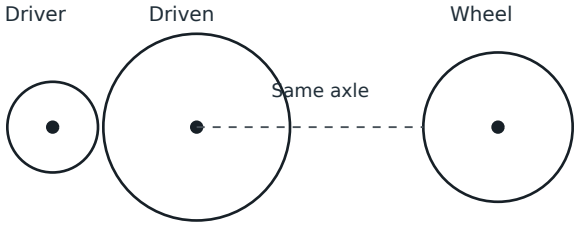


Travel per turn: ____ Ground speed: ____

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 12-tooth driver at 900 rpm turns a 36-tooth gear attached to wheels of radius 0.03 m. Find ideal wheel rpm and ground speed without slip.



Label teeth, rotation rates and wheel size.

Marzano check - circle: 1 2 3